

Procedural trees for real-time environments

Michał Piatkowski

Spelutveckling
Kandidat
30 hp
Fakulteten för teknik och samhälle
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Huvudhandledare: Georgios Palamas

Abstract—**REFINE**: Vegetation is a common part of procedurally generated environments. Level of detail techniques are used to reduce resource consumption based on distance without affecting perceivable quality. This thesis aims to explore applicable tree generation algorithms that could support level of detail for creating vast worlds in real-time. To this end, we present an optimized version of Space Colonization written in Rust.

Index Terms—Generative Tree Modeling, Models of Trees, Plant Growth, **TODO**:

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1. INTRODUCTION

TODO:

- Modeling of tree growth is useful for xyz...
- describing a volume of a structure or grid.
- these are advanced methods; mainly Neural Cellular Automata (NCA)
- sense-making: hot for modern systems and black-boxes

1.1. Motivation

REFINE: The first concept for this thesis went along the lines of “procedural noise-based terrain with Level of Detail (LOD)”, stemming from the lack of extreme view distances in real-time procedurally generated game worlds [src?]. Improving on this feature gap turned out vague even with basis in prior art [link No Man’s Sky, Minecraft, etc.]. Questions arose; Why is LOD not mainstream? Is there player interest in increased view distance? This doubt lead to an early narrowing in scope towards “procedural trees with LOD”.

Vegetation can make for effective decoration in making terrain varied and more interesting to players [src?]. Trees in particular play an important role due to their relative size in the player’s field of view, acting as subliminal landmarks [1]. We may exploit this through distinct mutations, with one of the most obvious variations being tree height when accounting for distance [src ...me? find science?].

TODO: 4 Questions (1,2,3 can essentially be answered with personal experience):

- 1) problem -
- 2) what others have done - billboards bad,
- 3) why the problem persists / ongoing research
- 4) what our research adds to this area (algos?)

TODO: review on LOD: <https://ieeexplore.ieee.org/abstract/document/1323963>

1.2. Problem

When generating procedural worlds for games, the designer tends to distribute resources and skills in a way that is most noticeable to the player **TODO**: src?. This tendency has a decisive effect in the context of visual features; nearby objects get more attention from the player and in turn the designer, while distant objects lose importance. In linear or closed-world games, the designer can deliberately choose where to allocate resources since the playable area is limited **TODO**: src?. Procedural generation greatly complicates such decisions, as we can no longer focus on specific parts, but have to instead produce a coherent whole. To cope with the vast search space, the designer intentionally imposes rules and constraints.

1.2.1. Grids: A fundamental and common restraint are grids that delimit space **TODO**: src?, which can be implicit like rooms in a dungeon, or explicit like individual voxels in terrain. Herein lies the tradeoff; our grid size is one of the deciding factors on how the designer spends their resources.

Take some of the possible features that can appear in an infinite sandbox game like terrain, biomes, vegetation, or structures **TODO**: src? – listed roughly from large to small. These need to be mixed in some way to make an engaging

experience, but in a procedural world, we don't know where the player will venture or the order in which they encounter features. We can guide them with specific cues, but it can be challenging to generate meaningful trails such as dirt paths leading to settlements, or landmarks that signal some nearby Point of Interest (POI) **TODO: src/example from cubeworld?**.

Hardware limitations are a driving force as well, where grids can be particularly versatile in accommodating memory limits, but also introduce parallelization opportunities.

TODO: section feels incomplete

1.3. Existing Solutions

Since the player viewport into the virtual world is limited, we can exploit various tricks to lighten the burden on both developers and hardware. Back in the day at the beginning of 3D graphics, Binary Space Partitioning (BSP) allowed us to render complex scenes faster by avoiding unnecessary work, mainly by painting only visible polygons front-to-back [2].

1.3.1. Level of Detail:

1.3.2. *Minecraft*: Space partitioning helps us with separation of concerns in the distributed system that is our game world **TODO: src?**. To better explain how one can utilize a hierarchy of grids, we use *Minecraft* as an example:

- 1) *Blocks* are the smallest unit i.e. the voxels.
- 2) *Sections* are portions containing 16^3 blocks.
- 3) *Chunks* are columns of sections – height can vary.
- 4) *Regions* are groups of 32×32 chunks.

World generation is done chunk-wise and over multiple consecutive steps, all on demand as chunks appear in the player's view distance [3]. Structure origins and biomes are calculated before any blocks can be placed. Terrain shaping follows, and is the most computationally expensive phase thanks to various density functions. The convenient property of these formulas is their continuity, given that most of them are multiple octaves of gradient noise with some linear arithmetic sprinkled in between. Density is deterministic and embarrassingly parallel, in other words.

Caves and cliffs are fun and all, but we have a concurrency problem right after, referred to as decoration; the various post-processing steps which place structures, vegetation, and ore – to name a few. The point of congestion is a possibility of decoration features intersecting multiple chunks, requiring well-defined order and dependency tracking to resolve.

TODO: section feels incomplete

1.3.3. *General Techniques*: **TODO: billboard, offline pre-processing, triangle decimation**

1.4. Ongoing Research

There is no solid example of a commercially successful voxel-based game as of yet that showcases extensive use of LOD for terrain in a way that meaningfully integrates with gameplay **TODO: src?**.

As a closely related genre, space exploration games like *Space Engineers* and *No Man's Sky* provide long view distance almost out of necessity. Striving for realistic scales without any sort of LOD is technically infeasible **TODO: src...?**. Without a preview or map of sorts **TODO: mention Twilight**

Forest map, the player would have a difficult time navigating planets, let alone choosing where to land their ship **TODO: src?**. Both games use hierarchies of voxels to store and represent terrain, albeit visualize them with *marching cubes* for a smooth look [4], compared to the blocky look people may be used to when they think of voxels **TODO: src that thought?**.

Getting back to the *Minecraft* scene, many recent technical achievements regarding LOD are modifications (mods) to the base game by passionate developers in the community. Working with a large existing codebase is difficult, and this is reflected in the way LOD is approached; popular mods, such as *Distant Horizons* and *Voxy* generate simplified geometry from the ground truth, which can be framed as a mix of view-dependent and hierarchical LOD [5]. The obvious drawback is how expensive and wasteful this process can be, generating fully detailed chunks when only a few upper sections happen to be visible.

1.4.1. *Novel Approaches*: <https://github.com/xandergos/terrain-diffusion-mc> **TODO: mention MC terrain diffusion mod?**

TODO: for Section 5: theory on how noise-based terrain could be inherently generated at different scales

TODO: mention MC sections and LOD mods

TODO: mention <https://veloren.net/>, apparently

TODO: mention ray/path-tracing and BVH acceleration? especially Nanite from UE5, and voxel-Lumen for foliage

1.5. Survey

Given the state of things, we can finally revisit our problem; why do procedurally generated games struggle to present the vastness of their worlds? This turns out to be a vast question in its own right **TODO: intertwine or merge with Section 1.1?**. A simple theory is that long view distances need too large of a technical investment for an unproven feature **TODO: sources?!**.

2. PRIOR WORK

TODO: intro paragraph:

- real-time generation for games
- not focused on realism, but resembles life (bending the rules)
- how did i manage to mention relativity... (subatomic scales are not needed)
- we don't need to simulate life to recreate it
 - ▶ this originated from the Space Colonization (SC) authors being proud of their algo "growing like actual trees"
 - ▶ structures generally have to fit the artistic vision of the given media to not feel out of place [src?].

2.0.1. Games:

Terraria: **TODO: Mention Terraria "Living Trees"? They were important inspiration after all...**

Minecraft: **TODO: Mention Minecraft mods that were used in thesis slides? Sounds fair to mention, considering that we can look at their code and see how large trees/structures**

were generated in practice. This gives us a good glimpse into upsides and pitfalls.

2.1. Growing Neural Gas

REFINE: Supervisor suggested checking out Growing Neural Gas (GNG) networks [6] when the initial goal was established. Even after exploring various GNG implementations, we concluded that GNG was not particularly relevant to the subject of growing trees, nor was it effective for real-time even with major optimizations [7].

Possible uses we came up with were related to creating bounding geometry for tree crowns. These could be used to spawn SC attractors (more on that later), or to construct meshes for rendering (which grew out of scope).

On the topic of GNG, we also explored the LOD aspect and found interesting applications for compressing point cloud colors [8], but without relevance to *growing* interesting trees.

2.2. Space Colonization

TODO: we can classify the methods, starting with SC: this is a space-filling algo, what else exists?

TODO:

- Algo and how it's good for trees [9]
- Shadow propagation based approach mentioned in [10]
- How environment/weather/light can be used to affect SC [11]

TODO:

- mention R* tree [12] (found by examining [13]) as critical optimization over other data-structures
- evaluate performance claims in [14]

2.3. Neural Cellular Automata

TODO: Explain the algo and how it may be an interesting avenue for vegetation [15]

TODO: Mention other attempts and uses outside of trees [16], [17]

2.4. Environmental Effects

TODO: Explain L-systems mentioned in [18]

TODO: Explain realistic tree models found in [19] and [10]

TODO: Explain how environment/weather/light affects tree growth [11]

3. METHODS

3.1. Rendering Framework

To ease experiments regarding our algorithms of choice, we develop a minimal framework with the primary purpose of visualizing point clouds, since most of the data we are working with is effectively represented by points.

The choice of tech stack was driven by the performance-oriented nature of this research, which is why we settled on Rust early and out of familiarity. This gave us access to high-quality libraries that provide essentials like graphics API (wgpu), immediate mode GUI (egui), and acceleration structures (rstar).

TODO: Write more?

3.2. Space Colonization

We implemented the SC algorithm from the ground up by following the method section for tree modeling [9]. Nearest Neighbor Search (NNS) is a fundamental part of the algorithm, and a $O(n^2)$ loop over the set of attractors is untenable.

3.2.1. Data Structures: Thus an octree library (oktree) was used as the first acceleration structure, getting us close to real-time generation.

Octrees served us well throughout development, but it became apparent that iterating over many large overlapping spheres was expensive when influence radii d_i on attractors was large. A naive workaround was to constrain the search to a certain radius **TODO: src?**, but this deviates us from the algorithm, in turn worsening growth dynamics by disregarding distant attractors.

Getting reminded that we are working with volumes and not just points, we pursued a spatial hash with cells that each referenced all intersecting attractors. These amortized lookups gave us a decent speedup, but increased both memory usage and removal time due to excessive duplication. Cost remained high even while utilizing indirect references (common in Rust to sidestep lifetime issues **TODO: src?**) as small as 16-bit; enough for 65535 attractors — 2^{16} indices minus 1 tombstone.

The structures we explored usually trade construction time and memory for improved lookup times. This was actually the reason for us looking beyond Voronoi diagrams in the first place, specifically with the expensive Delaunay triangulation step needed to construct them. An aspect we paid less attention to, but is worth noting, is tree balance and quality, which can vary at runtime as attractors are killed and nodes are spawned.

So far, neither octrees or spatial hashing proved effective against slowdown caused by smaller segment size D . At some point, our supervisor seemed to recognize this problem as related to DBSCAN [13], which led us to our final structure of choice: the R*-tree. Using an existing library (rstar), we managed to further decrease lookup time, trivialize construction cost while creating an optimal tree since we have all points upfront, all while maintaining fast NNS regardless of influence radius. This was now a functionally faithful reimplement of the SC algorithm.

3.2.2. Parameters: **TODO:** explain how parameters can be expanded upon, e.g. alternating node activations, non-immediate attractor death

The main variables d_i , d_k , and D can lead to very different generation times for a particular attractor cloud. Smaller segments lead to fewer attractors potentially being killed per new node, increasing the amount of iterations needed to consume all reachable attractors. Our choice of data structures may greatly alleviate the cost of large influence distances, but this variability can fundamentally not be eliminated without a different growth strategy and would likely require a new approach overall.

3.2.3. Attractor Spawning: **TODO:** describe how the cloud shapes the tree

TODO: Using Signed Distance Field (SDF) to spawn attractors

3.2.4. *Versatility:* With our definition of real-time **TODO:** define real-time in intro maybe?, we luckily do not need to worry about achieving predictable time complexities. We can tune both parameters and data, allowing designers to also allocate time towards post-processing of the generated skeleton. Point clouds are unlikely to be useful in their raw form, **TODO:** incorporate [19] ref but the methods for sprucing them up are vast.

The SC paper already mentions decimation and subdivision along branch curves, but these operate on points to improve detail. An example that is more relevant to us, and closer to the tail end of the content pipeline, is voxelization; the act of turning smooth geometry into a rigid grid **TODO:** any good reference?. **TODO:** shortly describe point->voxel process (and e.g. ->MC block)

3.2.5. *Incremental Growth:* **TODO:** SC can be grown over multiple steps (or over multiple game frames) allowing for massive structures without a hitch

3.3. Neural Cellular Automata

TODO: Describe the experiment around NCA [16]:

- Reproducing the open-source build found on GitHub by updating Python packages.
- Exploring the provided Jupyter notebook.
- Modifying the loss function to achieve different growth patterns. The original loss function is a combination of *Softmax* and *negative log likelihood* loss¹.

TODO:

- black box / related to evolutionary algo
- difficult for algo to settle on novel shapes

TODO:

- pruning methods; why is it important, how to measure?
- how easy is it to control appearance?
- run a test? benchmark?
- would like to see: insights around NCA loss function (helps reach a goal), not particularly a ready product

3.4. Persistent Homology

TODO: Measure/identify tree shape with “barcodes” [20]

4. RESULTS

4.1. Space Colonization

TODO: SC reimplementaion:

- We tested custom spatial hash buckets and octrees (slowest construction, unbalanced storage, unbalanced lookup time).

Elaborate on

- default vs. tuned insertion parameters for R* tree
- customized spatial hashing

TODO: Improve readability of these images...

and add more interesting images (probably as appendix)

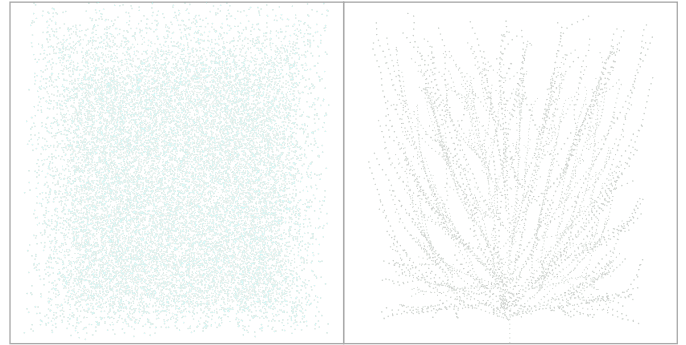


Figure 1. SC box before growth.

Figure 2. SC box after growth.

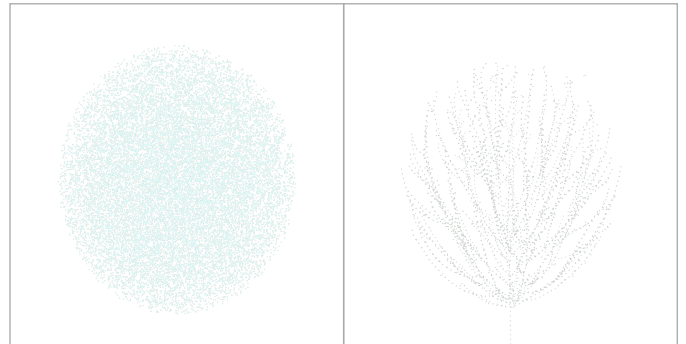


Figure 3. SC egg before growth.

Figure 4. SC egg after growth.

TODO:

- demonstrate parametrized examples from exploration
- variation comes from a combination of point cloud shape, influence distance, kill distance.

TODO: Write about data structure exploration and explain Table 1.

TODO: sift through git and get exact timings for buckets and octree, and maybe be more concise about the Memory column...

	Construct	Lookup	Memory
Octree	100ms	1500ms	Low
Spatial Hash	200ms	800ms	High
R*-tree	1.5ms	300ms	Low

Table 1. Data structure metrics for 20000 randomly distributed attractors.

TODO: Hardware:

- CPU: AMD Ryzen 7 7700X 8-Core Processor
- RAM: 2 × 32 GB, 5200 MT/s, CL32

4.2. Neural Cellular Automata

REFINE: Preliminary result around NCA [16]:

- Growth quickly collapses by scaling weights with values below one.
- Introducing more randomness to cell propagation creates spurious growths, which may be effective for tree variation, but may appear too close to unnatural overgrowth without introducing new restrictions.

¹<https://docs.pytorch.org/docs/stable/nn>



Figure 5. Original NCA tree

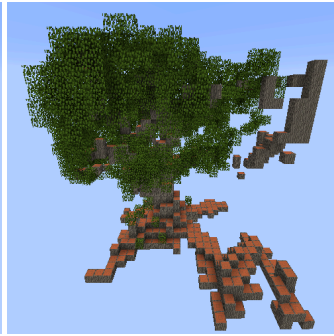


Figure 6. Experimental NCA tree with factors: mask = 0.2, fire = 0.75

- abstract triangles vs rigid cells (of NCA)
- genetic/neural algo to spawn SC attractors?

5. DISCUSSION

TODO:

- Compare our performance with original SC Voronoi structure (slowest construction, balanced storage, balanced lookup time).
- Compare our customizability with mentioned papers regarding tree modeling.

TODO: Describe how thesis progressed from a goal (generating trees in real-time for games) to exploring various prior work ([6], [9], [15])

TODO:

- What methods can be combined, and how?
- Loop back to LOD and reason around feasibility.

TODO:

- Compare SC vs shadow propagation [10]
- Light and shadow are key to creating nature-like trees.

TODO: Discuss future work from SC papers: convert scanned point clouds of real trees to algo params

TODO: Explain relevance of environment in tree growth:

- Local species (pests, competition, mycelium) may act as biomes to dictate tree attributes.

Generating a population of trees:

- quick poisson disc sample for initial placement?
- elaborate on how terrain can shape placement

6. FUTURE WORK

TODO: expand on Section 1.3; “generating noise at different LOD”

TODO: few insights into potential level-of-detail support (which was the initial motivation/goal).

Future: properly investigate the viability of these algorithms in a real-time environment, with the crux being that it should be scalable for large procedural worlds.

TODO: minimal rendering framework; the trees are not interesting/novel compared to prior work, and visualization of experiments was hindered by the framework.

Future: better rendering to

- 1) spend less time in debugging/testing
- 2) attempt reaching parity with [10] in particular

TODO: <https://github.com/peterbraden/genetic-lisa>

- related to [20] and

GLOSSARY

API	Application Programming Interface
BSP	Binary Space Partitioning
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
GNG	Growing Neural Gas
GUI	Graphical User Interface
LOD	Level of Detail
NCA	Neural Cellular Automata
NNS	Nearest Neighbor Search
POI	Point of Interest
SC	Space Colonization
SDF	Signed Distance Field

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